

YANZHOU, China

WMO: 549160

Lat: 35.5630N

Lon: 116.8424E

Elev: 174

StdP: 14.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	18.8	-4.5	4.4	31.5	0.0	5.5	29.7	15.6	37.9	13.9	35.7	2.8	90	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	94.4	76.7	92.3	76.8	90.2	76.2	83.0	89.5	81.7	88.1	80.2	86.4	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.4	164.4	87.3	79.9	156.3	86.1	78.5	148.9	84.5	47.4	89.5	45.8	88.5	44.2	86.6	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	12.5	10.9	10.7	98.5	2.7	2.3	8.7	100.1	7.1	101.4	5.6	102.7	3.7	104.3
			9.4	85.0	2.7	1.4	7.4	85.9	5.8	86.7	4.2	87.5	2.3	88.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	31.7	37.4	48.1	59.3	69.0	77.8	81.6	79.3	71.1	60.3	46.4	34.9
	DBStd	18.23	5.38	6.47	7.31	6.92	5.42	4.55	4.01	4.38	5.17	6.18	7.19	5.64
	HDD50	1674	567	353	124	8	0	0	0	0	0	4	150	468
	HDD65	4214	1032	772	524	195	23	0	0	0	9	169	557	933
	CDD50	4663	0	2	66	287	588	833	980	908	634	322	43	0
	CDD65	1729	0	0	1	24	146	384	515	443	193	23	0	0
	CDH74	17244	0	0	10	221	1378	4063	5710	4288	1411	162	1	0
	CDH80	6742	0	0	1	39	398	1789	2470	1635	393	17	0	0
	WSAvg	4.9	4.6	5.2	6.1	6.2	5.6	5.3	4.7	4.2	4.0	4.1	4.5	4.5
	PrecAvg	27.1	0.4	0.4	0.7	1.5	1.7	3.3	8.7	5.4	3.0	1.4	0.6	0.4
(15) Precipitation	PrecMax	46.4	1.6	1.5	2.4	6.5	5.4	8.5	15.9	14.4	7.2	6.0	3.9	2.0
	PrecMin	16.0	0.0	0.0	0.0	0.0	0.0	0.2	3.1	0.2	0.1	0.0	0.0	0.0
	PrecStd	7.1	0.4	0.4	0.6	1.5	1.5	2.2	3.9	3.3	1.9	1.3	0.7	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	61.6	75.1	84.5	92.1	97.4	97.1	94.5	90.7	82.4	70.8	55.1
		MCWB	40.7	49.7	59.7	68.7	69.2	71.7	78.7	82.5	74.1	64.1	57.7	47.4
	2%	DB	47.7	56.8	70.3	79.7	87.4	94.3	94.5	91.7	87.1	78.5	65.5	50.9
		MCWB	38.8	45.3	55.4	64.8	68.5	71.7	79.4	80.9	72.5	63.2	54.1	44.0
	5%	DB	44.6	53.2	65.8	76.0	84.3	91.8	92.6	89.7	84.4	75.4	62.0	47.8
		MCWB	36.9	42.8	53.0	62.5	67.3	71.2	79.8	79.2	70.5	61.8	52.4	41.2
	10%	DB	41.7	49.7	61.7	72.3	81.4	89.4	90.2	87.7	81.9	71.9	58.4	44.9
		MCWB	35.0	41.1	50.1	60.1	66.1	71.1	79.1	78.0	69.2	60.6	50.8	39.2
	0.4%	WB	43.8	52.1	62.0	72.0	73.9	80.1	85.1	85.0	79.7	69.3	60.0	49.5
		MCDB	48.2	59.0	71.5	81.2	82.9	88.5	91.1	91.0	85.6	74.4	66.3	52.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	47.4	57.7	67.7	71.9	77.9	83.4	83.1	76.1	66.5	57.2	45.8
		MCDB	45.3	54.5	67.2	76.2	81.9	86.5	90.2	89.1	82.5	74.1	62.5	49.2
	5%	WB	38.0	44.4	54.2	64.6	70.0	76.1	82.1	81.7	73.9	64.5	54.6	42.8
		MCDB	43.6	50.7	63.9	73.1	80.6	85.1	89.0	87.5	80.3	71.7	59.8	46.7
	10%	WB	35.7	41.9	51.3	61.9	68.2	74.3	80.8	80.3	72.1	62.5	51.7	40.1
		MCDB	40.5	48.5	60.1	70.4	78.2	83.5	87.7	85.6	78.3	69.0	57.3	43.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	17.9	19.2	19.8	20.3	19.1	14.0	14.2	17.8	19.3	18.1	16.8
		MCDBR	22.1	23.5	24.7	23.7	24.9	23.7	17.2	15.9	21.3	23.9	23.1	21.5
		MCWBR	14.2	14.1	13.0	11.9	10.0	7.3	5.7	6.3	7.7	10.1	12.4	14.0
	5% WB	MCDBR	19.1	19.9	22.7	21.7	20.6	17.3	14.4	13.4	16.9	17.8	18.6	17.6
		MCWBR	13.0	13.0	13.0	12.6	10.3	7.1	6.3	6.3	7.9	8.9	11.6	12.9
(40) Clear-Sky Solar Irradiance	taub		0.543	0.618	0.631	0.646	0.663	0.797	0.767	0.738	0.680	0.685	0.616	0.550
	taud		1.710	1.604	1.602	1.607	1.614	1.430	1.527	1.566	1.625	1.561	1.633	1.727
	Ebn at Noon		190	193	207	215	214	187	191	193	194	173	168	177
	Edh at Noon		58	73	79	82	83	99	90	85	76	75	63	55
(44) All-Sky Solar Radiation	RadAvg		726	899	1318	1570	1736	1653	1481	1407	1239	1014	770	671
	RadStd		86	133	107	107	123	139	136	170	135	105	108	78

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page